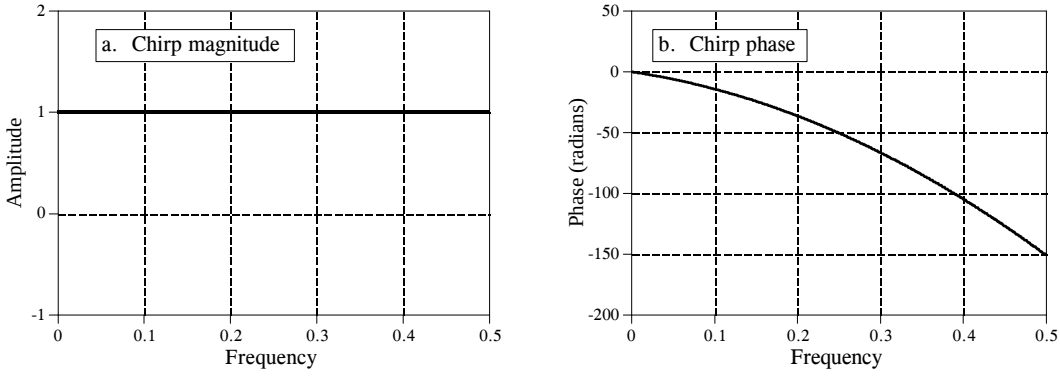

FIGURE 11-9
Frequency response of the chirp system. The magnitude is a constant, while the phase is a parabola.

The parameter α introduces a linear slope in the phase, that is, it simply shifts the impulse response left or right as desired. The parameter β controls the *curvature* of the phase. These two parameters must be chosen such that the phase at frequency 0.5 (i.e. $k = N/2$) is a multiple of 2π. Remember, whenever the phase is directly manipulated, frequency 0 and 0.5 must both have a phase of zero (or a multiple of 2π, which is the same thing).

Figure 11-10 shows an impulse entering a chirp system, and the impulse response exiting the system. The impulse response is an oscillatory burst that starts at a low frequency and changes to a high frequency as time progresses. This is called a *chirp* signal for a very simple reason: it sounds like the chirp of a bird when played through a speaker.

The key feature of the chirp system is that it is completely *reversible*. If you run the chirp signal through an *antichirp* system, the signal is again made into an impulse. This requires the antichirp system to have a magnitude of one, and the *opposite* phase of the chirp system. As discussed in the last

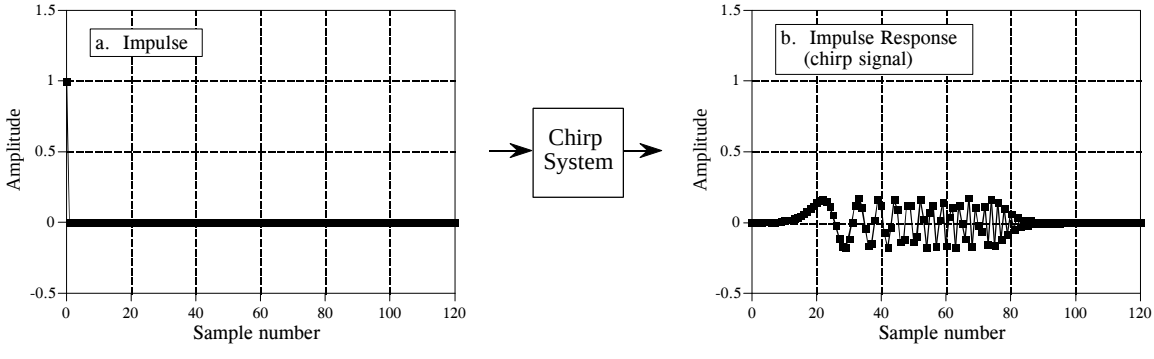

FIGURE 11-10
The chirp system. The impulse response of a chirp system is a chirp signal.